

One world, many playable places

DnDRom • Biomes and connected scene generation • September 8, 2026

Audience: DnDRom product and engineering. Scope: a local UE5 application targeting 8 GB VRAM and 16 GB RAM. This is the researched design for the next world-generation system. It distinguishes proposed work from the separately tested chunk-seam repair; it does not claim that towns, all biome families or ferries are already implemented.

The decision

Keep one persistent world identity. Let users and the local AI describe the places an adventure needs; let the procedural engine locate, connect and build them within that world. Detailed gameplay maps and distant scenery must be two representations of the same terrain, settlements and roads.

The current checkout already has eroded terrain, site selection and streamed scenery. Its campaign director still assigns a location-specific map seed, schematic map positions and abstract road links. Scene persistence stores separate map snapshots. Those are useful foundations, but they do not yet establish a shared geographic world or physically traversable connections. The next change should replace this geographic ownership model before adding more scene-specific terrain generators.

Microsoft's documented Minecraft pipeline generates terrain, assigns biomes using elevation and several environmental noise parameters, then adds structures and features. Heat and height capture only part of that process. This reference supports separating geography from settlement placement, rather than using a prompt to independently regenerate both. [Microsoft/Mojang, World Generation Overview, updated 2025-11-11](#).

What a harbor request should produce

For the island harbor, reserve an adequate dry settlement area and a connected navigable water body. Plan dock frontage, working/storage buildings, homes, a public gathering area and internal circulation. Reserve a mainland landing and a validated ferry route. A visible boat follows that route. The requested town cannot silently degrade to one building if the island is too small: expand the search, use another suitable shore, or expose the capacity conflict.

This proposal creates the physical world and travel structure first. NPC schedules, commerce, crowds and dynamic story simulation can then use it in the later “living world” phase.

A climate atlas above the render tiles

Generate environmental causes

Use stable world coordinates and versioned coarse geology to derive elevation, coast distance and prevailing winds. Cache a lower-resolution climate atlas containing mean temperature, seasonal amplitude, precipitation, aridity and soil moisture. Altitude affects temperature; wind transports moisture, and mountain barriers produce wetter windward and drier leeward regions.

AutoBiomes demonstrates coarse climate simulation, altitude-dependent temperature, moisture transport, precipitation and biome classification, followed by terrain detail and asset placement. It also blends biome borders. Its river and geological-layer simulation remained future work, and its climate timings are not evidence of real-time infinite-world performance. We should reuse its architecture while retaining DnDRom's erosion and explicit drainage ownership. [Fischer, Dittmann, Weller and Zachmann, AutoBiomes, 2020](#).

Our proposed initial scale is 8 km macroregions with 128 m climate cells; these are benchmark starting values. A 65 by 65 grid with eight float channels is about 132 KiB of raw data. This arithmetic excludes halos, caches, vegetation, textures, meshes and the AI model. Generate climate from shared upstream boundary conditions or a larger parent region, never an independent simulation in each render chunk.

Preserve hydraulic geography

The erosion system supplies drainage, slope, rock exposure and sediment. Store water-body IDs, basin levels, outlets and connectivity separately so a lake, ocean or river remains identifiable across scenes. A ship needs a connected navigable region with suitable depth, not simply a blue patch near a building.

McDonald's hydrology work uses particle motion, sediment transfer and stream history to form channels and meanders. The author removed the older lake flood-fill mechanism because it scaled poorly. Therefore this article alone does not solve a persistent infinite-world lake/ocean graph; that needs its own representation and validation. [Nicholas McDonald, Procedural Hydrology, 2023-12-12](#).

Bound the work

Prepare a coarse campaign region and index promising sites first. Refine terrain only for candidates and visible or soon-to-be-visited areas. Persist accepted results. Expensive regional work must run outside the render thread, support cancellation, and retain completed caches. Benchmark cold climate/erosion, cached lookup, meshing, upload and rendering separately.

Biomes as ecological mixtures

Support families and habitats

The proposed starting catalogue includes polar ice, tundra, boreal forest, temperate mixed forest, temperate rainforest, grassland/steppe, Mediterranean shrubland, hot desert, cold desert, savanna, tropical seasonal forest and tropical rainforest. These are product categories, not an assertion that ecology has one exhaustive finite taxonomy. NASA's biome overview provides a useful climate-and-community framing, but not a complete fantasy-world catalogue. [NASA Earth Observatory, Mission: Biomes, updated 2025-12-05](#).

Add habitat layers for alpine terrain, rocky slopes, rivers and riparian ground, wetlands, freshwater lakes, coasts and dunes, saline/marine areas, volcanic/geothermal terrain and underground spaces. A swamp requires suitable water and drainage; an alpine meadow depends on altitude and climate; a desert oasis can be a local wet habitat within a dry regional biome.

Treat enchanted, corrupted, fungal, crystalline and elemental environments as explicit fantasy overlays. They can change species, materials, atmosphere and declared physical rules. This keeps an intentional magical boundary distinguishable from an accidental abrupt climate transition.

Natural transitions and believable plants

Evaluate a small normalized mixture of compatible biome families at each world point. The same weights drive terrain layers, surface tint, plant species, density and distant canopy. Blend habitat suitability rather than spawning unrelated plant sets independently in adjacent tiles. Place canopy trees, understory and ground cover with competition, shade, soil, water and disturbance constraints.

Mojang's historical Java experiments specifically addressed fragmented biomes, temperature clashes, abrupt patches and climate-dependent mountain placement. They demonstrate why a plausible initial classifier still needs transition tests. They are historical design evidence, not an exact specification of current Java internals. [Henrik Kniberg/Mojang, World Generation in Java Available for Testing, 2021](#).

Make every supported family findable

Random noise does not guarantee a requested biome within a finite travel budget. At world creation, build a coarse index and verify that every supported family has a feasible region. If one is missing, reserve an appropriate unvisited macroregion using deterministic constraints before its detailed terrain exists. Keep those reservations in the world manifest. Never rewrite explored land or move accepted settlements to satisfy a later request. Unsupported habitats or impossible constraints must produce an explicit result instead of an unbounded search.

The local AI plans; the engine places

Persistent contracts

WorldManifest owns world ID, immutable seed, generator/climate/hydrology versions, region reservations, water bodies, accepted locations, regional styles and transport edges. A render tile is a cache address, not a world authority.

SceneRequest carries a stable request ID, gameplay purpose, environment constraints, settlement scale, mandatory roles, desired relations and allowed transport modes. A scene list can include an island harbor, mainland market, mountain fortress and forest encounter. The AI supplies semantic relationships such as “the warehouse is near the working dock”; it does not invent authoritative geographic coordinates.

LocatedScene stores world coordinates, reserved area, measured buildability, environmental identity, entrances and gameplay anchors. SettlementPlan stores parcels, buildings, streets, docks and stable building IDs. TransportEdge stores actual endpoint IDs, mode, route geometry, construction constraints and validation evidence. Runtime scene origins can move for rendering precision without changing world coordinates.

Holodeck found that direct LLM coordinate placement caused boundary and collision problems, and instead combined model-generated relations with constrained layout search. That supports our division of responsibility. Its experiments used GPT-4, indoor environments and an existing asset collection; they do not establish performance or quality for local outdoor generation. [Yang et al., Holodeck, CVPR 2024, revision 2024-04-22.](#)

Generation transaction

First normalize the requested scene list and separate hard constraints from preferences. Query the regional biome/site index, then measure candidate terrain, buildable area and water access. Reserve suitable sites and their connection corridors. Generate settlement parcels and transport geometry, validate them, then commit the accepted plan atomically. Failed placement should return the unsatisfied requirement and available alternatives while leaving the existing world intact.

Give the local model bounded tools such as querySites, proposeLocationProgram, connectLocations, validatePlan and revisePlan. Each returns compact measurements and explicit failure reasons. Persist tool results and plan versions; make retries idempotent. Preserve local speech/text entry and the app-owned model runtime. No external model service is required by this architecture.

Detailed buildings, enterable interiors and distant silhouettes must use the same building IDs and footprints. Interiors may load lazily, but doors and entrances must already agree with the exterior. A regional style changes construction grammar, materials, proportions and ornament; it must not silently replace the terrain or location seed.

Towns, bridges and ferry routes

Build a town program

Village-generation research combines terrain, water and vegetation maps with connections to the outside world. It alternates building placement and road growth, using interest functions for accessibility, water proximity, slope, clustering and defense. This is a better basis for a harbor town than scattering a fixed number of houses. Its examples are small European settlements, so broader architectural coverage requires additional grammar packs. [Emilien et al., Procedural Generation of Villages on Arbitrary Terrains, 2012.](#)

Our proposed harbor acceptance fixture uses a town-scale dry site, multiple residential parcels, working/storage buildings, a social center, a public waterfront, an accessible dock and a mainland destination. Exact counts should be editable product parameters and constrained by available land. Keep roads, entrances and dock approaches clear before filling the remaining space with foliage. Every required role must either exist or have a reported placement failure.

Connect real endpoints

Road search should penalize excessive grade, tight turns, water and expensive earthworks, while favoring useful existing routes. Smooth the result and revalidate its grade and clearance afterward. Generate the roadbed from its centerline with controlled crossfall; do not inherit arbitrary terrain roll. Galin's road research supports terrain-aware routing with slope, curvature and crossing constraints. Detailed bridge claims in this review rely on the author PDF's indexed extraction because full retrieval failed. [Galin et al., Procedural Generation of Roads, 2010.](#)

Our proposed bridge catalogue includes timber beam/trestle, stone slab/arch and rope/suspension families. Select only feasible spans and approaches before applying regional style. Validate dry support sites, deck grade, clearance and water obstruction. A stylistic request cannot make an unsupported structure geometrically valid.

Ferries are a separate routing mode. Both docks must reach the same connected navigable water body; the route must maintain vessel draft, shoreline clearance and turning room. Persist the resulting route and drive a boat along it using distance-based movement and dock arrival states. This is our extension to the transport system, not a capability established by the road paper. Full passengers, schedules and economic activity remain later work.

Architecture belongs to regions

Keep cultural/artistic style separate from climate. Regional material supply, construction tradition, wealth, age and local hazards can influence a grammar pack. Climate alone does not imply a particular culture. Start with explicitly supported packs, parameter variation and reference-driven material styling; record unsupported features instead of claiming arbitrary historical architecture is solved.

Implementation order and acceptance

First preserve continuity

The current seam patch changes sampling and surface ownership, not the accepted wave equations. It shares terrain derivatives across authored chunks, shares water contours and run-up data, and removes overlapping coarse water from the local region. Matching boundary positions alone is insufficient: normals, material weights, water levels and feature masks must agree too. Geometry-clipmap research likewise treats geometry and shading transitions as explicit parts of terrain continuity. [Asirvatham and Hoppe, Terrain Rendering Using GPU-Based Geometry Clipmaps, GPU Gems 2, 2005.](#)

Next implement the versioned WorldManifest and migrate scene coordinates without changing existing campaign terrain. Preserve legacy snapshots when the original world cannot be reconstructed faithfully. Then add the climate/habitat atlas and bounded biome index, followed by settlement programs, transport routing, regional building grammars and shared near/far settlement representations. Finish this phase with the island-harbor-to-mainland-ferry scenario before adding broader living-world simulation.

Epic's PCG runtime generation supports distance-based scheduling and hierarchical generation grids. That is useful scheduling precedent, but changing frameworks alone would not solve DnDRom's persistent world-plan requirements. Stream accepted plans and generate detail near relevant sources; do not invent a new location each time a tile loads. [Epic Games, Using PCG Generation Modes, UE5.8 documentation, accessed 2026-09-08.](#)

Required gates

Continuity: shared world samples agree across chunks, LODs, generation order and save/load. Compare fixed-camera water and terrain captures, and retain the accepted foam/motion checks. Do not equate a clean runtime log with visual acceptance.

Ecology: each supported biome family has an indexed feasible site; climate transitions are gradual except declared magic; wet habitats have hydrological support; plants respect water, slope, soils and clearings. Validate windward/leeward behavior on controlled terrain.

Gameplay: every mandatory scene is connected; building entrances reach circulation; ordinary foundations are dry; bridges have valid approaches; ferries remain navigable; adding a scene preserves existing geography and IDs. Distant town outlines must match their playable plans.

Performance: measure cold and cached generation separately, report CPU/GPU upload stalls and memory, and repeat on the target 8 GB VRAM / 16 GB RAM machine with local AI active. The cited work does not prove this app meets that target. These gates are proposed acceptance criteria, not completed tests of the future system.

The research is sufficient to choose this architecture. Remaining uncertainties are engineering benchmarks, local-model constraint-following quality, the breadth of supported style assets, and habitat-specific rendering work. Those need prototypes and measured acceptance, rather than additional broad searches or promises of unlimited fidelity.